12TH CHEMISTRY · CHAPTER 30 · CHAPTER TEST

Spectroscopy-2 NMR

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 30	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. NMR is based on:

(a) Bond vibration (b) Ion mass
(c) Electron jumps (d) Nuclear spin

2. The chemical shift of TMS is:

(a) 1.0 ppm (b) 0.0 ppm
(c) 7.0 ppm (d) 12 ppm

3. A proton with 3 neighbours appears as a:

(a) Quartet (b) Doublet
(c) Triplet (d) Singlet

4. The solvent used in proton NMR is:

(a) Water (b) Ethanol
(c) Deuterated chloroform (d) Benzene

5. Carbon-13 peaks of propane are:

(a) 1 (b) 2
(c) 3 (d) 4

6. Peak area in proton NMR shows the:

(a) Shift (b) Solvent
(c) Number of protons (d) Spin

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Why are deuterated solvents used in proton NMR?
ii. State two reasons why TMS is used as a standard.
iii. Define chemical shift.
iv. State the n+1 rule.
v. Why do O-H protons appear as a singlet?
vi. How many carbon-13 peaks does butane show? Why?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain the principle of proton NMR.
(b) Predict the proton NMR spectrum of chloroethane.

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